

Quantum Field Theory

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Syllabus (Two Semesters) — AGD Edition, verified against the table of contents

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Course Information

- **Primary text:** Abrikosov, Gor'kov, and Dzyaloshinski, *Quantum Field Theoretical Methods in Statistical Physics*, 2nd ed. (Pergamon, 1965). This syllabus reads the book cover to cover, in order, as the sole text.
- **Prerequisites:** undergraduate quantum mechanics, classical mechanics, statistical mechanics, and comfort with complex analysis and Fourier transforms.
- **Duration:** 27 weeks. Semester A (Weeks 1–13) covers Chapters I–III: many-particle systems at low temperature through the full zero- and finite-temperature Green's function formalism (§1–§17). Semester B (Weeks 14–27) covers Chapters IV–VIII: Fermi liquid theory, interacting bosons, electromagnetic radiation in absorbing media, superconductivity, and the transport equation (§18–§42).
- **Reading key:** [AGD §X] denotes Abrikosov, Gor'kov, and Dzyaloshinski, with subsections noted where the book itself numbers them.

1 Semester A: Many-Particle Systems and the Green's Function Formalism

1.1 Phase 1: Properties of Many-Particle Systems at Low Temperatures (Chapter I)

This phase covers Weeks 1–3.

1.1.1 Week 1: Elementary Excitations and Fermi Liquids

- AGD §1 introduces quasiparticles and works out the energy spectrum of liquid He^4 : the Bose-liquid spectrum and superfluidity. §2 introduces Fermi liquids: excitations, quasiparticle energy, and sound.

Application. The quasiparticle is the single idea that makes strongly interacting many-body systems tractable; everything later in the book refines this picture.

Readings. [AGD §1–2]

1.1.2 Week 2: Second Quantization and the Dilute Bose Gas

- AGD §3 develops second quantization for bosons and fermions. §4 applies it to the dilute Bose gas, a fully worked example of an interacting many-body system.

Application. The dilute Bose gas is the simplest non-trivial interacting system and a template for every diagrammatic calculation to come.

Readings. [AGD §3–4]

1.1.3 Week 3: The Dilute Fermi Gas

- AGD §5 treats the dilute Fermi gas, the fermionic counterpart of Week 2's calculation.

Application. Comparing the Bose and Fermi dilute gases highlights how statistics alone reshapes the ground state and excitation spectrum.

Readings. [AGD §5]

Checkpoint. Checkpoint: be comfortable with quasiparticles, second quantization, and the qualitative difference between dilute interacting Bose and Fermi gases. This closes Chapter I.

1.2 Phase 2: Quantum Field Theoretical Methods at $T = 0$ (Chapter II)

This phase covers Weeks 4–7.

1.2.1 Week 4: The Interaction Representation and the Green Function

- AGD §6 introduces the interaction representation. §7 defines the single-particle Green function, its free-particle form, analytic properties, the physical meaning of its poles, and its generalization to a system in an external field.

Application. The poles of the Green function in frequency are the energies of the system's elementary excitations — the same quasiparticles introduced in Chapter I, now given a precise field-theoretic definition.

Readings. [AGD §6–7]

1.2.2 Week 5: Wick's Theorem and Feynman Diagrams

- AGD §8 covers the basic principles of diagram techniques: changing independent variable from N to μ , Wick's theorem, and Feynman diagrams.

Application. Wick's theorem is the combinatorial engine that turns the perturbation series into a sum of diagrams.

Readings. [AGD §8]

1.2.3 Week 6: Diagram Rules in Coordinate and Momentum Space

- AGD §9 works out the rules for drawing diagrams for different types of interaction, with worked examples in both coordinate space and momentum space.

Application. Learning to read diagrams off the page, in both representations, is the central diagrammatic skill of the course.

Readings. [AGD §9]

1.2.4 Week 7: Dyson's Equation, Vertex Parts, and Ground State Energy

- AGD §10 sums diagrams into Dyson's equation, introduces the vertex part and many-particle Green functions, and computes the ground-state energy diagrammatically.

Application. Dyson's equation resums an infinite diagram series into a single closed equation for the dressed propagator.

Readings. [AGD §10]

Checkpoint. Checkpoint: derive Dyson's equation, define the vertex part, and explain how the ground-state energy is extracted from the diagram series. This closes Chapter II.

1.3 Phase 3: The Diagram Technique at Finite Temperatures (Chapter III)

This phase covers Weeks 8–13 and completes the Green's function formalism.

1.3.1 Week 8: Temperature-Dependent Green Functions

- AGD §11 introduces temperature-dependent (Matsubara) Green functions: their general properties and the free-particle case.

Application. Imaginary time turns thermal averages into a structure formally parallel to the $T = 0$ technique just developed.

Readings. [AGD §11]

1.3.2 Week 9: Finite-Temperature Perturbation Theory

- AGD §12 develops perturbation theory at finite temperature: the interaction representation and Wick's theorem in this setting.

Application. The same Wick-theorem machinery from Chapter II reappears, now organized around Matsubara frequencies.

Readings. [AGD §12]

1.3.3 Week 10: The Finite-Temperature Diagram Technique

- AGD §13 develops the diagram technique in coordinate space with worked examples. §14 transforms this to the momentum representation, again with worked examples.

Application. Coordinate- and momentum-space diagrams are two languages for the same physics; fluency in both pays off in every later chapter.

Readings. [AGD §13–14]

1.3.4 Week 11: The Thermodynamic Potential from Diagrams

- AGD §15 derives the perturbation theory series for the thermodynamic potential directly from the diagram technique.

Application. Diagrams are not just for excitation spectra: they also generate the equilibrium thermodynamics of an interacting system.

Readings. [AGD §15]

1.3.5 Week 12: Dyson’s Equation at Finite Temperature

- AGD §16 derives the finite-temperature Dyson equation and many-particle Green functions, and connects the Green functions to the thermodynamic potential Ω .

Application. This is the finite-temperature counterpart of Week 7’s Dyson equation, now tied directly to thermodynamics.

Readings. [AGD §16]

1.3.6 Week 13: Analytic Properties of Finite-Temperature Green Functions

- AGD §17 studies the real-time Green functions G at finite temperature and their analytic properties, completing the bridge between the imaginary-time formalism and physically measurable response functions.

Application. Analyticity arguments let you extract retarded/advanced response functions directly from the imaginary-time formalism.

Readings. [AGD §17]

Checkpoint. Midterm checkpoint: the full Green’s function formalism — zero-temperature and finite-temperature, diagrams, Dyson’s equation, and analytic structure — is now complete (§1–§17). This closes Semester A.

2 Semester B: Fermi Liquids, Bosons, Radiation, Superconductivity, and Transport

2.1 Phase 4: Fermi Liquid Theory (Chapter IV)

This phase covers Weeks 14–17.

2.1.1 Week 14: The Vertex Part and Zero Sound

- AGD §18 studies the properties of the vertex part for small momentum transfer and derives zero sound.

Application. Zero sound is a direct experimental signature of Fermi-liquid quasiparticles, observed in liquid ^3He .

Readings. [AGD §18]

2.1.2 Week 15: Effective Mass and Specific Heat

- AGD §19 derives the effective mass, the connection between the Fermi momentum and the number of particles, Bose branches of the spectrum, specific heat, and the damping of quasiparticles in a Fermi liquid.

Application. This section is the technical heart of Landau Fermi liquid theory: every later thermodynamic and transport result builds on it.

Readings. [AGD §19]

2.1.3 Week 16: Small Total Momentum and Electron–Phonon Interactions

- AGD §20 studies special properties of the vertex part when the total momentum of colliding particles is small. §21 develops electron–phonon interactions: the vertex part, the phonon Green function, the electron Green function, and the correction to the linear term in electronic specific heat.

Application. Electron–phonon coupling is the bridge between idealized Fermi liquid theory and real metals.

Readings. [AGD §20–21]

2.1.4 Week 17: A Degenerate Plasma

- AGD §22 applies the Fermi liquid formalism to a degenerate plasma: the vertex part for small momentum transfer, the electron spectrum, and thermodynamic functions.

Application. The degenerate electron plasma is a key application of Fermi liquid methods to real conductors.

Readings. [AGD §22]

Checkpoint. Checkpoint: explain the Landau quasiparticle picture, derive the zero-sound dispersion qualitatively, and connect effective mass to specific heat. This closes Chapter IV.

2.2 Phase 5: System of Interacting Bosons (Chapter V)

This phase covers Weeks 18–20.

2.2.1 Week 18: Bosons at Absolute Zero and Their Green Functions

- AGD §23 applies field theory methods to a system of bosons at absolute zero. §24 develops the Green functions: the structure of the equations, their analytic properties, and their behavior for small momenta.

Application. The interacting Bose gas requires its own (Gor’kov-style) Green function structure, foreshadowing the superconductor calculation later in the course.

Readings. [AGD §23–24]

2.2.2 Week 19: The Dilute Non-Ideal Bose Gas, Revisited Diagrammatically

- AGD §25 returns to the dilute non-ideal Bose gas with the full diagram technique: the connection between chemical potential and self-energy, the low-density approximation, the effective interaction potential, and the resulting Green functions and spectrum.

Application. This redoes Week 2’s calculation with the full machinery of Chapters II–III, showing how much sharper the diagrammatic approach is.

Readings. [AGD §25]

2.2.3 Week 20: The Spectrum Near Threshold and Finite-Temperature Bosons

- AGD §26 studies the single-particle excitation spectrum near its end-point: behavior near the phonon-creation threshold and near two-excitation breakup thresholds. §27 extends the formalism to interacting bosons at finite temperature.

Application. These threshold phenomena are subtle, exactly-solvable test cases for the analytic structure developed in Chapter III.

Readings. [AGD §26–27]

Checkpoint. Checkpoint: explain how the Bose-gas Green function differs structurally from the Fermi-gas case, and why. This closes Chapter V.

2.3 Phase 6: Electromagnetic Radiation in an Absorbing Medium (Chapter VI)

This phase covers Weeks 21–22.

2.3.1 Week 21: Radiation Green Functions and the Dielectric Constant

- AGD §28 develops radiation Green functions in an absorbing medium. §29 calculates the dielectric constant from them.

Application. The dielectric response of a real material is, microscopically, a Green function calculation — the same formalism as Chapters II–III, now applied to photons.

Readings. [AGD §28–29]

2.3.2 Week 22: Van der Waals and Molecular Interaction Forces

- AGD §30 derives Van der Waals forces in a non-uniform dielectric. §31 extends this to molecular interaction forces between solids, between atoms in solutions, and in thin films on a solid surface.

Application. Van der Waals forces, normally introduced as a chemistry curiosity, fall directly out of the same Green function technique used for excitation spectra.

Readings. [AGD §30–31]

Checkpoint. Checkpoint: derive the dielectric constant from the radiation Green function for a simple model. This closes Chapter VI.

2.4 Phase 7: Theory of Superconductivity (Chapter VII)

This phase covers Weeks 23–26.

2.4.1 Week 23: Model and the Cooper Instability

- AGD §32 introduces the general model and interaction Hamiltonian for superconductivity. §33 derives the Cooper phenomenon: the instability of the non-interacting Fermi sea to arbitrarily weak attraction, via the equation for the vertex part, and determines the transition temperature.

Application. The Cooper instability shows that an arbitrarily weak attractive interaction destabilizes the normal Fermi liquid at low temperature.

Readings. [AGD §32–33]

2.4.2 Week 24: Fundamental Equations and the Phonon Model

- AGD §34 derives the system of fundamental equations for a superconductor at absolute zero, in an external electromagnetic field (gauge invariance), and at finite temperature. §35 deduces these equations specifically in the phonon model.

Application. This is the complete microscopic (Gor'kov) derivation of the gap equation, recovered entirely from Green's functions and diagrams.

Readings. [AGD §34–35]

2.4.3 Week 25: Thermodynamics and Weak Electromagnetic Fields

- AGD §36 computes the thermodynamics of superconductors: temperature dependence of the gap, specific heat, and critical field. §37 treats superconductors in a weak electromagnetic field, both constant and variable.

Application. These sections connect the microscopic gap equation to directly measurable thermodynamic and electromagnetic quantities.

Readings. [AGD §36–37]

2.4.4 Week 26: Near T_c and Superconducting Alloys

- AGD §38 studies the properties of superconductors close to the transition temperature in an arbitrary magnetic field. §39 develops the theory of superconducting alloys: residual resistance of normal metals and electromagnetic properties of superconducting alloys.

Application. Real superconductors are alloys with disorder; this section is where the clean BCS picture meets realistic materials.

Readings. [AGD §38–39]

Checkpoint. Checkpoint: derive the BCS gap equation, explain gauge invariance of the superconducting equations, and state how disorder modifies the clean-limit results. This closes Chapter VII.

2.5 Phase 8: Transport Equation for Excitations in a Fermi Liquid (Chapter VIII)

This phase covers Week 27 and closes the course.

2.5.1 Week 27: Non-Equilibrium Fermi Liquids, the Vertex Part, and Synthesis

- AGD §40 introduces non-equilibrium properties of a Fermi liquid and states the transport problem. §41 studies the analytical properties of the vertex part needed to solve it. §42 derives the equation for the vertex part and the resulting transport equation, closing the book. Use the rest of the week to review the full conceptual arc from second quantization to the transport equation.

Application. The transport equation shows that even far from equilibrium, the Fermi-liquid quasiparticle picture — built from Green functions, diagrams, and vertex parts across the whole course — still governs the dynamics.

Readings. [AGD §40–42]

Theorem 1 (Dyson’s Equation). *The full (dressed) single-particle Green function G is related to the free Green function G_0 and the self-energy Σ (the sum of one-particle-irreducible diagrams) by*

$$G^{-1} = G_0^{-1} - \Sigma.$$

This resums an infinite diagrammatic series into a single closed equation, and holds in identical form at $T = 0$ (§10) and at finite temperature (§16).

3 Final Assessment

- Write a 4–6 page expository essay deriving and explaining one result in detail, drawn entirely from AGD.
- Suggested topics include the BCS gap equation, the zero-sound dispersion relation, the dielectric constant from the radiation Green function, or the Fermi-liquid transport equation.

4 Recommended & Further Readings

- Fetter and Walecka, *Quantum Theory of Many-Particle Systems*: a complementary many-body Green function text, similar in spirit to AGD.